

NT-C-005 | Power BI | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Business Intelligence
Target learners	Students and professionals who need dashboarding, reporting and business-intelligence skills.
Prerequisites	Basic spreadsheet knowledge; SQL is helpful.
Approved duration	Approved working range: 8-12 weeks / 80-120 guided hours.
Final outcome	Prepare data, build a reliable semantic model, write DAX and publish secure, usable Power BI reports.
Assessment profile	Data/AI

Approved tools / platforms

Power BI Desktop, Power Query, DAX, Power BI Service, Excel/CSV/SQL data sources.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-005 | Power BI | NT-C-005-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Power BI workflow, requirements, KPIs; Sources, Power Query, profiling, cleaning
Completion evidence	Dashboard specification; Prepared query set
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - BI orientation

Objective: Power BI workflow, requirements, KPIs

Observable evidence: Dashboard specification

Concept	Approved knowledge statement	Application behaviour	Evidence
Power BI workflow	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when power bi workflow is appropriate; apply it in a Power BI task; verify the result.	Dashboard specification
requirements	Documented stakeholder needs, constraints and acceptance conditions. In Power BI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when requirements is appropriate; apply it in a Power BI task; verify the result.	Dashboard specification
KPIs	A defined metric tied to a business objective and decision. In Power BI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when kpis is appropriate; apply it in a Power BI task; verify the result.	Dashboard specification

Module 2 - Connect & prepare

Objective: Sources, Power Query, profiling, cleaning

Observable evidence: Prepared query set

Concept	Approved knowledge statement	Application behaviour	Evidence
Sources	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sources is appropriate; apply it in a Power BI task; verify the result.	Prepared query set
Power Query	A repeatable data connection and transformation layer. In Power BI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when power query is appropriate; apply it in a Power BI task; verify the result.	Prepared query set
profiling	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when profiling is appropriate; apply it in a Power BI task; verify the result.	Prepared query set
cleaning	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cleaning is appropriate; apply it in a Power BI task; verify the result.	Prepared query set

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Dashboard specification; Prepared query set, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-005 | Power BI | NT-C-005-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Star schema, relationships, dimensions; Measures, filter context, iterators
Completion evidence	Validated semantic model; Measure library
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - Data modelling

Objective: Star schema, relationships, dimensions

Observable evidence: Validated semantic model

Concept	Approved knowledge statement	Application behaviour	Evidence
Star schema	A dimensional model with fact tables connected to descriptive dimension tables. In Power BI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when star schema is appropriate; apply it in a Power BI task; verify the result.	Validated semantic model
relationships	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when relationships is appropriate; apply it in a Power BI task; verify the result.	Validated semantic model
dimensions	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dimensions is appropriate; apply it in a Power BI task; verify the result.	Validated semantic model

Module 4 - DAX foundation

Objective: Measures, filter context, iterators

Observable evidence: Measure library

Concept	Approved knowledge statement	Application behaviour	Evidence
Measures	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when measures is appropriate; apply it in a Power BI task; verify the result.	Measure library
filter context	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when filter context is appropriate; apply it in a Power BI task; verify the result.	Measure library
iterators	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when iterators is appropriate; apply it in a Power BI task; verify the result.	Measure library

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Validated semantic model; Measure library, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-005 | Power BI | NT-C-005-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Time intelligence, variables, calculation patterns; Visual choice, interactions, accessibility
Completion evidence	Business KPI model; Interactive report
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Advanced DAX

Objective: Time intelligence, variables, calculation patterns

Observable evidence: Business KPI model

Concept	Approved knowledge statement	Application behaviour	Evidence
Time intelligence	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when time intelligence is appropriate; apply it in a Power BI task; verify the result.	Business KPI model
variables	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when variables is appropriate; apply it in a Power BI task; verify the result.	Business KPI model
calculation patterns	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when calculation patterns is appropriate; apply it in a Power BI task; verify the result.	Business KPI model

Module 6 - Report design

Objective: Visual choice, interactions, accessibility

Observable evidence: Interactive report

Concept	Approved knowledge statement	Application behaviour	Evidence
Visual choice	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when visual choice is appropriate; apply it in a Power BI task; verify the result.	Interactive report
interactions	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when interactions is appropriate; apply it in a Power BI task; verify the result.	Interactive report
accessibility	Design and implementation that allows people with diverse abilities to use a product. In Power BI, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when accessibility is appropriate; apply it in a Power BI task; verify the result.	Interactive report

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Business KPI model; Interactive report, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-005 | Power BI | NT-C-005-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Publish, refresh, workspace, sharing; RLS, optimisation, governance, capstone
Completion evidence	Published report; Secure optimised dashboard
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Power BI Service

Objective: Publish, refresh, workspace, sharing

Observable evidence: Published report

Concept	Approved knowledge statement	Application behaviour	Evidence
Publish	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when publish is appropriate; apply it in a Power BI task; verify the result.	Published report
refresh	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when refresh is appropriate; apply it in a Power BI task; verify the result.	Published report
workspace	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when workspace is appropriate; apply it in a Power BI task; verify the result.	Published report
sharing	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sharing is appropriate; apply it in a Power BI task; verify the result.	Published report

Module 8 - Security & performance

Objective: RLS, optimisation, governance, capstone

Observable evidence: Secure optimised dashboard

Concept	Approved knowledge statement	Application behaviour	Evidence
RLS	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when rls is appropriate; apply it in a Power BI task; verify the result.	Secure optimised dashboard
optimisation	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when optimisation is appropriate; apply it in a Power BI task; verify the result.	Secure optimised dashboard
governance	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when governance is appropriate; apply it in a Power BI task; verify the result.	Secure optimised dashboard
capstone	A defined competency within Power BI. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when capstone is appropriate; apply it in a Power BI task; verify the result.	Secure optimised dashboard

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Published report; Secure optimised dashboard, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-005 | Power BI | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-005-SK01	BI orientation	Prerequisite foundation	Dashboard specification	Critical
NT-C-005-SK02	Connect & prepare	M01 plus earlier foundations	Prepared query set	Critical
NT-C-005-SK03	Data modelling	M02 plus earlier foundations	Validated semantic model	Supporting
NT-C-005-SK04	DAX foundation	M03 plus earlier foundations	Measure library	Supporting
NT-C-005-SK05	Advanced DAX	M04 plus earlier foundations	Business KPI model	Supporting
NT-C-005-SK06	Report design	M05 plus earlier foundations	Interactive report	Critical
NT-C-005-SK07	Power BI Service	M06 plus earlier foundations	Published report	Supporting
NT-C-005-SK08	Security & performance	M07 plus earlier foundations	Secure optimised dashboard	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-005 | Power BI | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-005-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of BI orientation and Connect & prepare?	The learner should provide Dashboard specification; Prepared query set and explain how the work applies Power BI workflow, requirements, KPIs; Sources, Power Query, profiling, cleaning. Recall alone is insufficient.
NT-C-005-S01-EX02	Scenario/remediation	A learner reports completing Connect & prepare but cannot produce Prepared query set. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-005-S01, request Prepared query set, and allow a new equivalent test only after remediation.
NT-C-005-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Data modelling and DAX foundation?	The learner should provide Validated semantic model; Measure library and explain how the work applies Star schema, relationships, dimensions; Measures, filter context, iterators. Recall alone is insufficient.
NT-C-005-S02-EX02	Scenario/remediation	A learner reports completing DAX foundation but cannot produce Measure library. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-005-S02, request Measure library, and allow a new equivalent test only after remediation.
NT-C-005-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Advanced DAX and Report design?	The learner should provide Business KPI model; Interactive report and explain how the work applies Time intelligence, variables, calculation patterns; Visual choice, interactions, accessibility. Recall alone is insufficient.
NT-C-005-S03-EX02	Scenario/remediation	A learner reports completing Report design but cannot produce Interactive report. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-005-S03, request Interactive report, and allow a new equivalent test only after remediation.
NT-C-005-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Power BI Service and Security & performance?	The learner should provide Published report; Secure optimised dashboard and explain how the work applies Publish, refresh, workspace, sharing; RLS, optimisation, governance, capstone. Recall alone is insufficient.
NT-C-005-S04-EX02	Scenario/remediation	A learner reports completing Security & performance but cannot produce Secure optimised dashboard. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-005-S04, request Secure optimised dashboard, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-005 | Power BI | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Sales KPI dashboard	Power Query, model, DAX	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Admissions funnel report	Relationships and interactions	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Management performance dashboard	RLS, refresh, documentation	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-005 | Power BI | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Power BI Developer - Trainee	Power Query 34% [critical]; modelling 33% [critical]; DAX 33%	Published dashboard	If a critical skill is below 60 or published dashboard is missing, return Improve First or Needs Evidence with the linked module/project.
2	BI Analyst - Junior	KPIs 34% [critical]; DAX 33% [critical]; storytelling 33%	Business report and analysis	If a critical skill is below 60 or business report and analysis is missing, return Improve First or Needs Evidence with the linked module/project.
3	Reporting Analyst	Data preparation 34% [critical]; refresh 33% [critical]; visualisation 33%	Repeatable reporting solution	If a critical skill is below 60 or repeatable reporting solution is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-005 | Power BI | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Power BI for?	Students and professionals who need dashboarding, reporting and business-intelligence skills.
What should I know before starting?	Basic spreadsheet knowledge; SQL is helpful.
How long is the approved learning journey?	Approved working range: 8-12 weeks / 80-120 guided hours.
What will I be able to demonstrate?	Prepare data, build a reliable semantic model, write DAX and publish secure, usable Power BI reports.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Power BI knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

course_code=NT-C-005; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12